



Durée: 2h

## Calcul Intégral

### Exercice I

Calculer les intégrales suivantes :

1)  $\int_0^{2\pi} \cos^2(x) dx.$

2)  $\int_0^1 \frac{dx}{\sqrt{1-x^2}}.$

3)  $\int_0^1 \frac{x}{x^4+1} dx.$

4)  $\int_e^{e^2} \frac{1}{x(\ln x)^2} dx.$

### Exercice II

A l'aide d'une intégration par partie calculer les intégrales suivantes :

1)  $I_1 = \int_1^e \ln(t) dt.$

2)  $I_2 = \int_0^1 \arctan(t) dt.$

3)  $I_3 = \int_0^{\frac{\pi}{2}} t \sin^3(t) dt.$

4)  $I_4 = \int_0^{\frac{\pi}{2}} e^t \cos(t) dt.$

### Exercice III

1) Décomposer en éléments simple les fractions rationnelles :

$$f(x) = \frac{x+1}{(x^2+1)(x-2)} \text{ et } g(x) = \frac{1}{x(x-1)}.$$

2) Calculer

$$\int_0^1 f(x) dx \text{ et } \int_2^3 g(x) dx.$$

### Exercice IV

Calculer les intégrales suivantes en utilisant le changement de variable précisé :

1)  $A = \int_1^e \frac{\ln t}{t} dt$ , en posant  $u = \ln(t)$ .

2)  $B = \int_0^1 \sqrt{1-t^2} dt$ , en posant  $t = \cos(t)$ .

3)  $C = \int_0^1 \frac{t^3}{\sqrt{t+1}} dt$ , en posant  $u = \sqrt{t+1}$ .



Exo 2 :

$$1) \int_0^{2\pi} \cos^2(x) dx$$

$$\left[ \frac{x + \sin x \cos x}{2} \right]_0^{2\pi}$$

$$= \frac{4\pi}{2} = 2\pi$$

$$2) \int_0^1 \frac{dx}{\sqrt{1-x^2}}$$

$$\int_0^1 \frac{1}{\sqrt{1-x^2}} dx$$

$$\left[ \arcsin(x) \right]_0^1$$

$$= \frac{\pi}{2}$$

$$3) \int_0^1 \frac{x}{x^4+1} dx$$

on pose  $u = x^2$   
 $du = 2x dx$   
 $dx = \frac{du}{2x}$

Si  $x=0$   $u=0$   
 $x=1$   $u=1$

$$\int_0^1 \frac{x}{u^2+1} \frac{du}{2x}$$

$$\frac{1}{2} \int_0^1 \frac{1}{u^2+1} du$$

$$\frac{1}{2} \left[ \arctan(u) \right]_0^1$$

$$= \frac{1}{2} \frac{\pi}{4}$$

$$= \frac{\pi}{8}$$



$$4) \int_e^{e^2} \frac{1}{x \ln^2 x} dx$$

$$\int_e^{e^2} \frac{\frac{1}{x}}{\ln^2(x)} dx$$

$$\left[ \frac{-1}{\ln x} \right]_e^{e^2}$$

$$\frac{-1}{\ln e^2} + \frac{1}{\ln e}$$

$$-\frac{1}{2} + 1 = \frac{1}{2}$$

Exo 2 :

$$I_1 = \int_1^e \ln(t) dt$$

$$u = \ln(t) \quad u' = 1$$

$$u' = \frac{1}{t} \quad v = t$$

$$\left[ t \ln(t) \right]_1^e - \int_1^e t \cdot \frac{1}{t} dt$$

$$\left[ t \ln(t) \right]_1^e - \int_1^e 1 dt$$

$$\left[ t \ln t \right]_1^e - \left[ t \right]_1^e$$

$$e \ln e - \ln 1 - e + 1$$

$$e - e + 1$$

$$I_1 = 1$$

$$I_2 = \int_0^1 \arctan(t) dt$$

$$u = \arctan(t) \quad u' = 1$$

$$u' = \frac{1}{1+t^2} \quad v = t$$

$$\left[ t \arctan(t) \right]_0^1 - \int_0^1 \frac{t}{1+t^2} dt$$

$$\left[ t \arctan(t) \right]_0^1 - \frac{1}{2} \int_0^1 \frac{2t}{1+t^2} dt$$

$$\left[ t \arctan(t) \right]_0^1 - \frac{1}{2} \left[ \ln(1+t^2) \right]_0^1$$

$$\frac{\pi}{4} - \frac{1}{2} \ln(2)$$



$$\int_0^{\frac{\pi}{2}} t \sin^3(t) dt$$

$$u = t \quad v' = \sin(t) - \sin(t) \cos^2(t)$$

$$u' = 1 \quad v = -\cos(t) - \frac{1}{3} \cos^3(t)$$

$$\left[ -t \left( \cos t + \frac{1}{3} \cos^3(t) \right) \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos(t) + \frac{1}{3} \cos^3(t)$$

$$\int_0^{\frac{\pi}{2}} \cos(t) + \frac{1}{3} (\cos(1 - \sin^2(t)))$$

$$\left[ -t \left( \cos(t) + \frac{1}{3} \cos^3(t) \right) \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos(t) + \frac{1}{3} \cos(t) - \frac{1}{3} \cos(t) \sin^2$$

$$\left[ -t \left( \cos(t) + \frac{1}{3} \cos^3(t) \right) \right]_0^{\frac{\pi}{2}} + \left[ \sin t + \frac{1}{3} \sin t - \frac{1}{3} \times \frac{1}{3} \sin^3 t \right]_0^{\frac{\pi}{2}}$$

$$I_4 = \int_0^{\frac{\pi}{2}} e^t \cos(t) dt$$

$$u = e^t \quad v' = \cos(t)$$

$$u' = e^t \quad v = t + \sin t$$

$$\left[ t e^t \sin t \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} e^t \sin t$$

$$\int_0^{\frac{\pi}{2}} e^t \sin t$$

$$u = e^t \quad v' = \cos(t)$$

$$u' = e^t \quad v = -\sin t$$

$$\left[ -e^t \cos t \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} -e^t \cos t dt$$

$$\int_0^{\frac{\pi}{2}} e^t \cos t dt = \left[ e^t \sin t \right]_0^{\frac{\pi}{2}} + \left[ e^t \cos t \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} e^t \cos t dt$$

$$2 \int_0^{\frac{\pi}{2}} e^t \cos t dt = \left[ e^t \sin t \right]_0^{\frac{\pi}{2}} + \left[ e^t \cos t \right]_0^{\frac{\pi}{2}}$$

$$= \left[ e^t \sin t \right]_0^{\frac{\pi}{2}} + \left[ e^t \cos t \right]_0^{\frac{\pi}{2}}$$

$$\frac{e^{\frac{\pi}{2}} - 1}{2} = \frac{e^{\frac{\pi}{2}}}{2} - \frac{1}{2}$$



3 :

$$f(x) = \frac{x+1}{(x^2+1)(x-21)}$$

$$\frac{x+1}{(x^2+1)(x-21)} = \frac{ax+b}{x^2+1} + \frac{c}{x-21}$$

On multiple par  $(x-21)$

$$\frac{(x+1)(x-21)}{(x^2+1)(x-21)} = \frac{(ax+b)(x-21)}{x^2+1} + \frac{c(x-21)}{x-21}$$

Pour  $x = 21$

$$\frac{22}{442} =$$

On multiple par  $x$

$$\frac{x^2+x}{(x^2+1)(x-21)} = \frac{ax^2+bx}{x^2+1} + \frac{Cx}{x-21}$$

Pour  $\frac{1}{x^2+x}$

$$\frac{x^2}{x^3} = \frac{ax^2}{x^2} + \frac{Cx}{x}$$

$$\frac{1}{x} = a + c$$

$$0 = a+c \Rightarrow c = -a \Rightarrow a = -\frac{22}{442}$$



$$x = 0$$

$$\frac{1}{-21} = b + \frac{c}{-21}$$

$$b = \frac{1}{-21} + \frac{c}{21}$$

$$b = \frac{c-1}{21}$$

$$b = \frac{-22}{442}$$

$$21$$

$$b = \frac{-232}{4641}$$

$$\frac{x+1}{(x^2+1)(x-21)} = \frac{\frac{-22}{442}x - \frac{232}{4641}}{x^2+1} + \frac{\frac{22}{442}}{x-21}$$

$$\int_2^3 \frac{x+1}{(x^2+1)(x-21)} dx = \int_2^3 \frac{\frac{-22}{442}x - \frac{232}{4641}}{x^2+1} dx + \int_2^3 \frac{\frac{22}{442}}{x-21} dx$$

$$\int_2^3 \left( \frac{\frac{-22}{442}x}{x^2+1} - \frac{\frac{232}{4641}}{x^2+1} + \frac{\frac{22}{442}}{x-21} \right) dx$$

$$\frac{-22}{442} \int_2^3 \frac{2u du}{u^2+1} - \frac{232}{4641} \int_2^3 \frac{1}{u^2+1} du + \frac{22}{442} \int_2^3 \frac{1}{u-21} du$$

$$\frac{-22}{442} \left[ \ln|x^2+1| \right]_2^3 - \frac{232}{4641} \left[ \arctan(x) \right]_2^3 + \frac{22}{442} \left[ \ln|x-21| \right]_2^3$$



$$f(x) = \frac{1}{x(x-1)}$$

$$\frac{1}{x(x-1)} = \frac{a}{x} + \frac{b}{x-1}$$

On multiplie par  $(x-1)$

$$\frac{x-1}{x(x-1)} = \frac{a(x-1)}{x} + \frac{b(x-1)}{(x-1)}$$

Pour  $x = 1$

$$1 = b$$

On multiplie par  $x$

$$\frac{x}{x(x-1)} = \frac{ax}{x} + \frac{bx}{x-1}$$

Pour  $x \rightarrow \infty$

$$\frac{x}{x^2} = \frac{ax}{x} + \frac{bx}{x}$$

$$\frac{1}{x} = a + b$$

$$0 = a + b$$

$$a = -b$$

$$a = -1$$

$$g(x) = \frac{-1}{x} + \frac{1}{x-1}$$

3

$$\int_2^3 \frac{1}{x(x-1)} dx = \int_2^3 \left( \frac{-1}{x} + \frac{1}{x-1} \right) dx$$

$$= - \left[ \ln|x| \right]_2^3 + \left[ \ln|x-1| \right]_2^3$$

$$= -\ln 3 + \ln 2 + \ln 2 - \ln 1$$

$$= 2\ln 2 - \ln 3$$

Exo4 :

$$A = \int_1^e \frac{\ln(t)}{t} dt$$

$$u = \ln(t)$$

$$du = \frac{1}{t} dt$$

$$dt = t du$$

$$\text{Si } t=1$$

$$u=0$$

on sait que

$$\text{Si } t=e$$

$$u=1$$

$$\sin^2(u) + \cos^2(u) = 1$$

$$\sin^2 = 1 - \cos^2 u$$

$$\sin u = \sqrt{1 - \cos^2 u}$$

$$\int_0^1 \frac{u}{t} du$$

$$\int_0^1 u du$$

$$\left[ \frac{1}{2} u^2 \right]_0^1 = \frac{1}{2}$$

$$A = \frac{1}{2}$$

$$B = \int_0^1 \sqrt{1-t^2} dt$$

$$t = \cos(u)$$

$$dt = -\sin(u) du$$

$$t=0$$

$$u = \frac{\pi}{2}$$

$$t=1$$

$$u=0$$

$$-\int_{\frac{\pi}{2}}^0$$

$$\sqrt{1-\cos^2(u)} \sin u du$$

$$\sin^2(u) + \cos^2(u) = 1$$

$$\sin^2 = 1 - \cos^2 u$$

$$\sin u = \sqrt{1 - \cos^2 u}$$

$$\sin u = \sqrt{1 - \cos^2 u}$$

$$\int_0^{\frac{\pi}{2}} \sin u \sin u du$$

$$\int_0^{\frac{\pi}{2}} \sin^2 u du$$

$$\left[ \frac{2u - \sin(2u)}{4} \right]_0^{\frac{\pi}{2}} = \frac{\pi}{4}$$

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$$\left[ \frac{2u - \sin(2u)}{4} \right]_0^{\frac{\pi}{2}} = \frac{\pi}{4}$$

$$B = \frac{\pi}{4}$$



$$= \int_0^1 \frac{t^3}{\sqrt{t+1}} dt$$

$$u = \sqrt{t+1}$$

$$du = \frac{1}{2\sqrt{t+1}} dt$$

$$dt = 2\sqrt{t+1} du$$

$$t=0 \quad u=1$$

$$t=1 \quad u=\sqrt{2}$$

$$\int_1^{\sqrt{2}} \frac{t^3}{x} 2u du$$

$$u = \sqrt{t+1}$$

$$u^2 = t+1$$

$$t = u^2 - 1$$

$$\int_1^{\sqrt{2}} t^3 du$$

$$= \int_1^{\sqrt{2}} (u^2 - 1)^3 du$$

$$(u^2 - 1)^3 = u^6 - 3u^4 + 3u^2 - 1$$

$$= \int_1^{\sqrt{2}} (u^6 - 3u^4 + 3u^2 - 1) du$$

$$= \left[ \frac{1}{7} u^7 \right]_1^{\sqrt{2}} - 3 \left[ \frac{1}{5} u^5 \right]_1^{\sqrt{2}} + 3 \left[ \frac{1}{3} u^3 \right]_1^{\sqrt{2}} - \left[ u \right]_1^{\sqrt{2}}$$

Fin | Liman